Cloud Computing Tutorial - 4 and 5

CloudSim Simulation

Observation:

Tutorial 4 and 5

Q. What is cloudsim?

sol: - cloudsim is a simulation tool that helps researchers and students study how cloud computing environments work without needing real cloud infrastructure. It is used to model and test cloud computing systems, including virtual machines (VMs), data centers, and scheduling policies, before deploying them in real-world scenarios.

Q. What does cloudsim do?

sol: cloudsim allows users to:

1. Simulate cloud infrastructure
2. Testing scheduling & Resource allocation
3. Evaluate performance
4. Optimize cloud services

Cloudsim is a framework for simulating cloud computing environments and evaluating different cloud strategies. It is implemented in Java and requires Java and eclipse for execution.

Prerequisites: - JDK 8 or higher
- Eclipse IDE
- cloudsim source code
- Apache Commons Math Library

Add cloudsim source code to eclipse
- Manually copy & paste code into eclipse project
1. open eclipse and create a new java project
2. paste cloudsim source code
3. refresh the project

Add required libraries
- Add external JAR files

Run a cloudSim Example

Eg: CloudSimExample1.java in source folder

output will display in console

Output:

Starting CloudSimExample1...

Initializing...

Starting Cloudsim version 3.0...

Datacenter_0 is starting...

⋮

simulation completed

output status → success

CloudSimExample1 finished!

CloudSim Simulation:

```
C:\Users\Admin>java --version
java 23.0.1 2024-10-15
Java(TM) SE Runtime Environment (build 23.0.1+11-39)
Java HotSpot(TM) 64-Bit Server VM (build 23.0.1+11-39, mixed mode, sharing)

C:\Users\Admin>
```

Fig 1.1: JDK installation verification

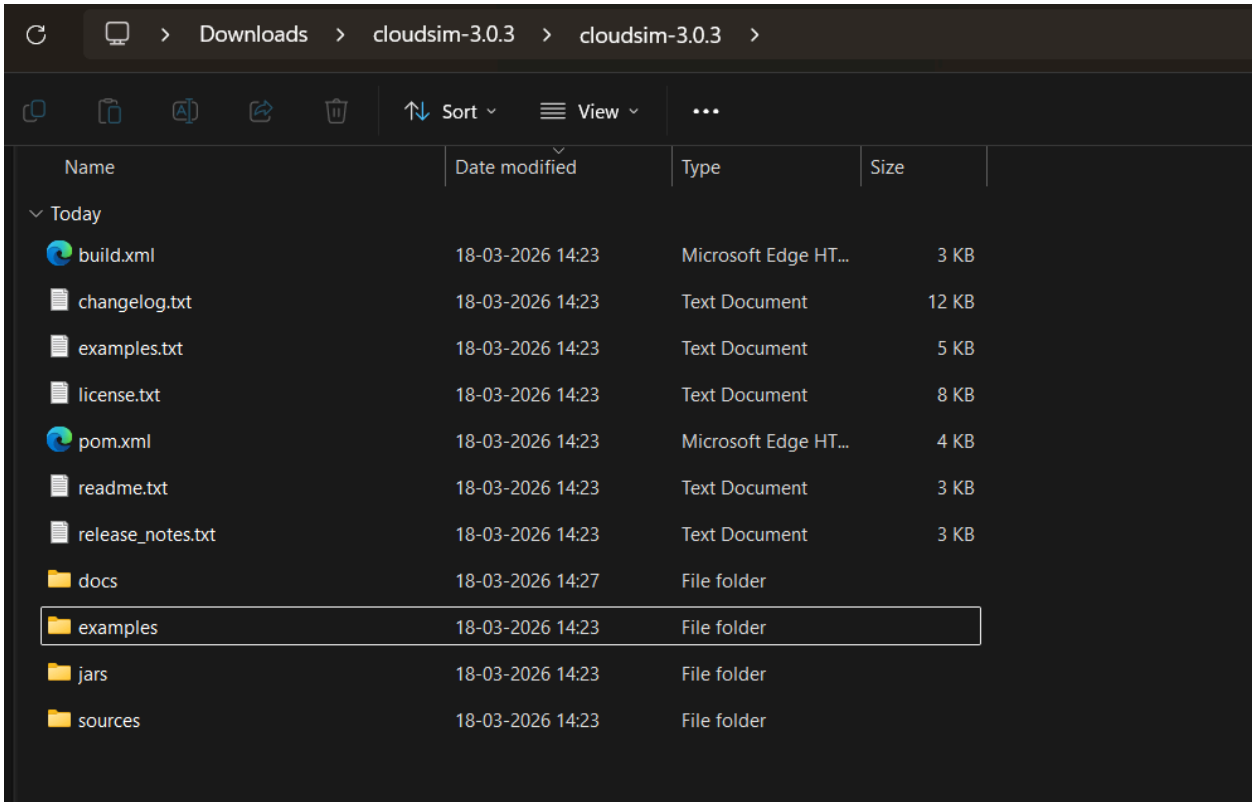

Fig 1.2: cloudsim zip folder extraction

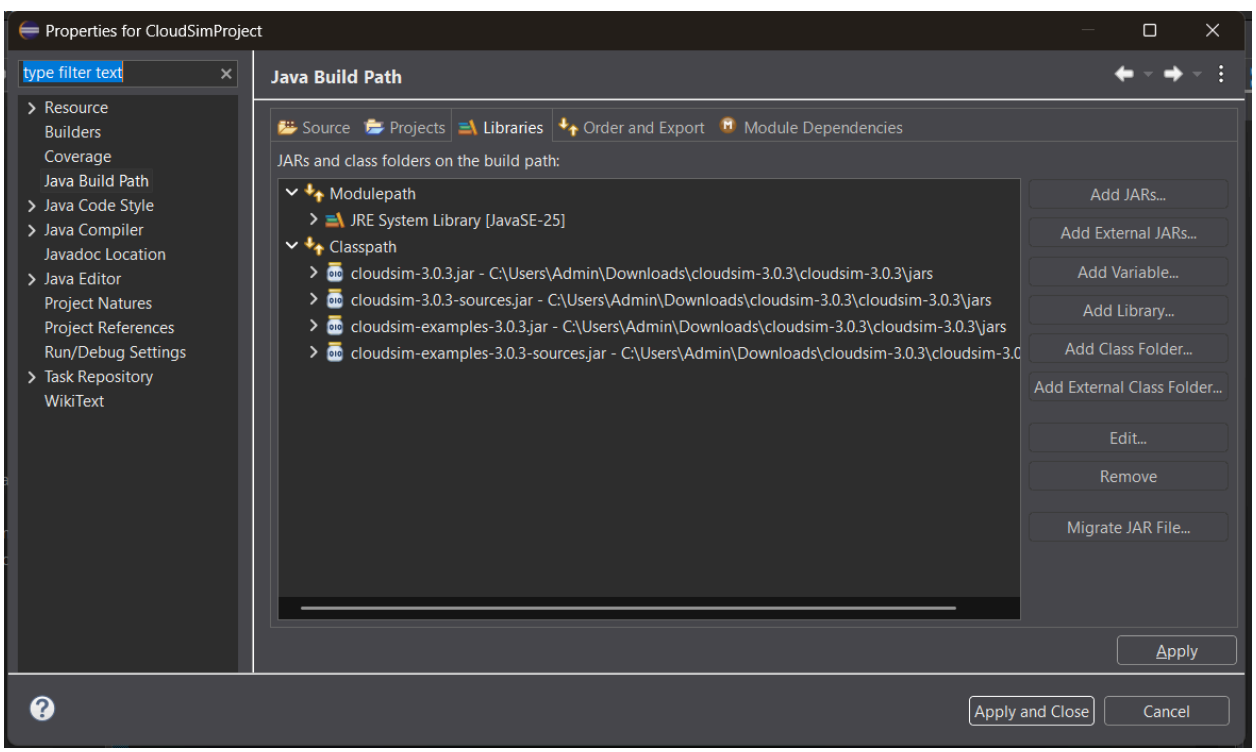

Fig 1.3: CloudSimProject properties

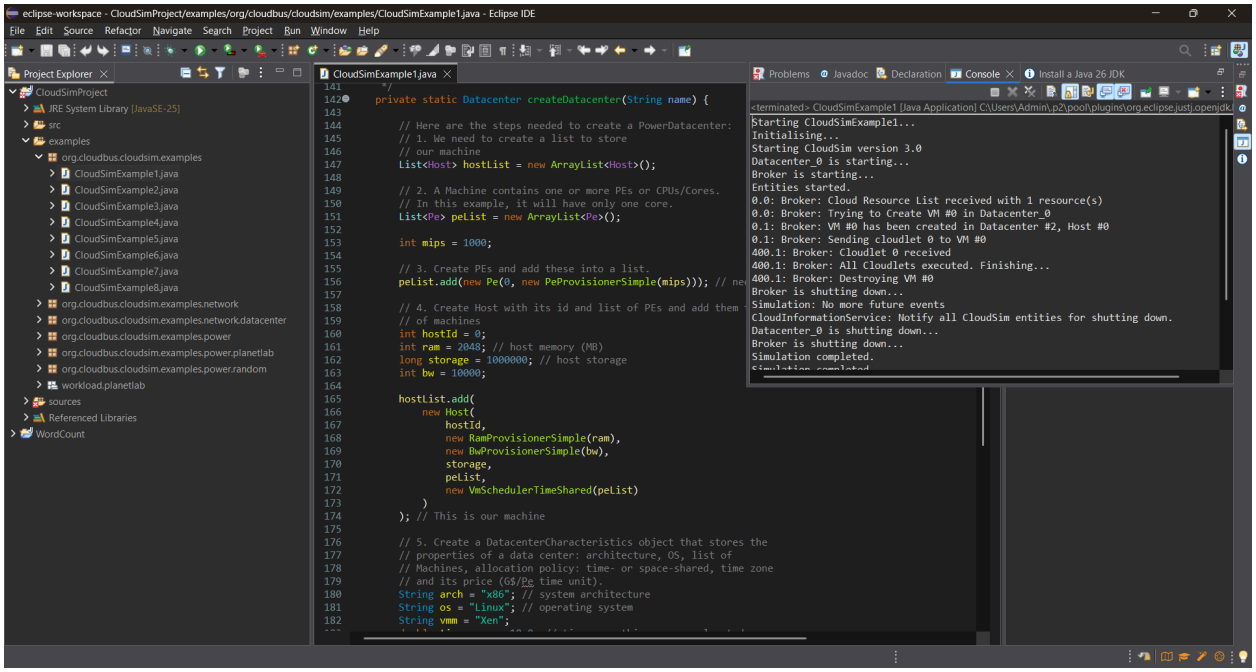

Fig 1.4: Running CloudSimProject - CloudSimExample1.java program

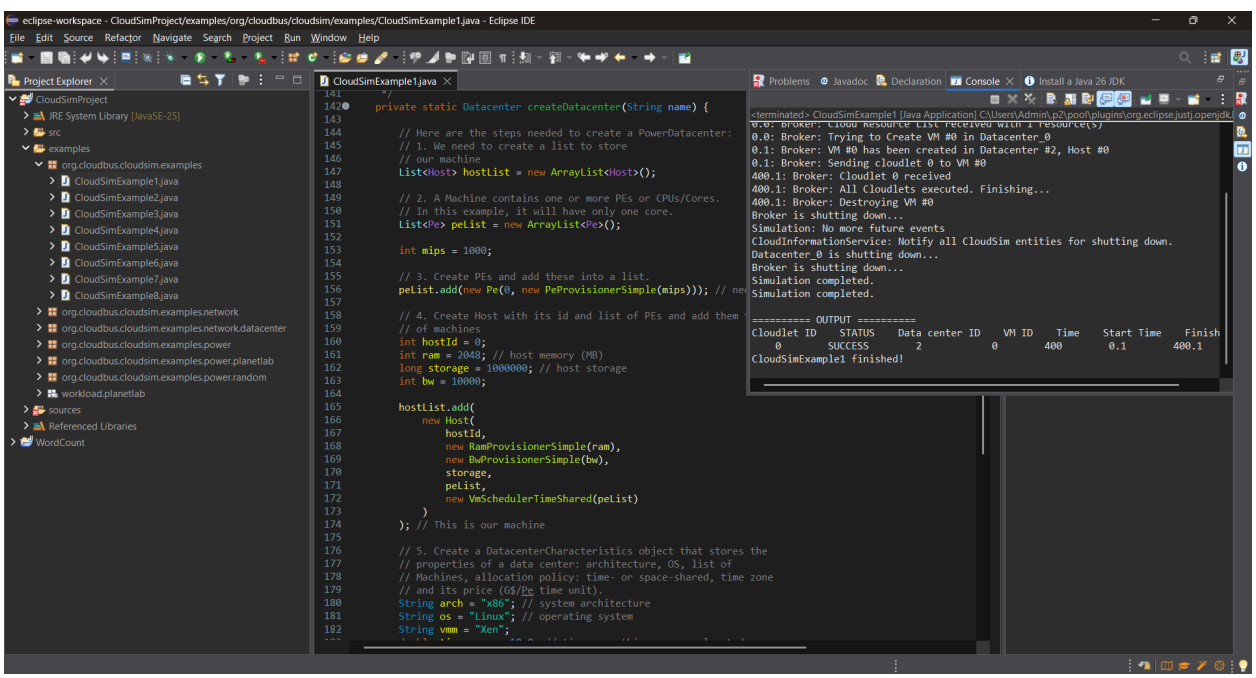

Fig 1.5: Simulation successful